

**Recognizing Social Faux Pas and Indirect Requests in Chinese Contexts: A Study on
Large Language Models**

CSS5120 Computational Linguistics

ZHANG Xuyang 225030243, LIU Xiaolin 225030214

1. Introduction

1.1 Problem Statement: The Cultural Divergence in Cognitive Mapping

Theory of Mind (ToM) represents the capacity to attribute mental states such as beliefs and intentions to others, which is the cornerstone of social intelligence. However, social cognitive abilities are highly culture dependent. This study addresses a critical gap: the performance disparity between Western-centric and Chinese-native LLMs in navigating high-context social dynamics. Specifically, we investigate whether models trained primarily on Western corpora (e.g., GPT-4o) suffer from a "high-context cognitive impairment" compared to those with native Chinese training (e.g., DeepSeek, ERNIE) when processing "subtext" and "face-saving" (*Mianzi*) mechanisms. The core question is whether "native cultural DNA" provides a deeper intuitive understanding of social faux pas than the "translated logic" of Western-centric models.

1.2 The Significance of Cultural Alignment

The problem of Western-centric bias in AI is not merely technical but societal. As Tao et al. (2024) stated, LLMs frequently favor the values of English-speaking nations, which can lead to non-Western users encountering misunderstandings or offense. This study matters because these subtle individual cognitive biases can accumulate over time, potentially reshaping broader cultural systems. Understanding the comparative strengths of Chinese versus Western models is essential for developing AI that preserves regional cultural authenticity and ensures equitable global AI deployment.

1.3 Literature Review and Research Gaps

Existing research has established a foundational but incomplete map of AI social cognition:

- **General ToM Baselines:** Strachan et al. (2024) demonstrated that GPT-4 can exhibit human-level performance in general social tasks. However, this "human-like" performance is largely validated within Western cognitive paradigms.
- **Cross-Cultural Bottlenecks:** The SAGE benchmark (Guo et al., 2025) indicates that LLMs often lag in scenario reasoning that requires deep cultural mapping.
- **Alignment Failures:** Kharchenko et al. (2025) found that models struggle to maintain local values like collectivism when generating responses.
- **Research Gap:** Current literature frequently treats "cultural bias" as a static attribute. There is a lack of comparative research on the dynamic adaptability of different model architectures when exposed to incremental levels of cultural knowledge injection.

1.4 Innovation and Contributions: A Comparative Framework

Building upon the work of Strachan et al. (2024) and the SAGE benchmark (Guo et al., 2025), this project introduces a specialized evaluation framework to quantify the cultural gap between Western and Chinese LLMs. Our innovations include:

1. **Dual-Model Comparison:** Unlike previous studies focusing on single model families, we directly compare Western benchmark models (GPT-4o) with native Chinese models (ERNIE, DeepSeek, Qwen) to distinguish between "internalized patterns" and "extrinsic reasoning".
2. **Quantifying the "Learning Curve":** By using a Three-Stage Injection Protocol (Zero, Weak, and Strong), we measure the Conceptual Sensitivity of Chinese models versus the Information Integration Capacity of Western models.

3. **Cross-Culture Core Concepts (CCCs):** We implement the CCC framework to evaluate how models handle "high-context logic" versus "low-context logic," filling the gap in ToM studies within non-Western social contexts.

2. Methodology

2.1 Evaluation Framework and Metrics

To quantify the Large Language Models' (LLMs) proficiency in navigating Chinese high-context social nuances, we employ a multi-tiered evaluation framework based on the SAGE benchmark (Guo et al., 2025). The evaluation systematically compares four models (Western model: GPT-4o; Native Chinese models: DeepSeek-V3, Qwen-Max, and ERNIE-4.0) across 50 culturally dense scenarios. We will calculate the Standardized Categorical Accuracy, and the results will be visualized using a Grouped Bar Chart comparing the accuracy of each model.

2.1.1 Scoring Standardization

Given the distinct scoring scales for the two primary tasks, we employ Standardized Categorical Accuracy (Acc_c) to ensure cross-task comparability:

- Task A: Indirect Requests (0-2 scoring): Measures the model's ability to map literal statements to implicit actions within high-context hierarchies.
- Task B: Social Faux Pas (0-3 scoring): Measures the model's sensitivity to "Face-saving" and interpersonal harmony.

The calculation formula is as follows:

$$ACC_c = \frac{\sum Scores_{observed}}{\sum Scores_{max}}$$

2.1.2 Three-Stage Injection Protocol

To observe the "learning curve" of cultural intelligence, we implement a longitudinal experimental design consisting of three incremental prompt injection levels:

- Zero Injection (ACC_{zero}): Baseline performance without cultural prompts.
- Weak Injection (ACC_{weak}): Providing a brief definition of cultural concepts (e.g., the concept of "Mianzi").
- Strong Injection (ACC_{strong}): Providing detailed reasoning chains and cultural background examples.

2.1.3 Performance Gain Metrics (SAGE-based Analysis)

Beyond static accuracy, we define three Differential Gain Metrics to analyze the models' cognitive adaptability:

- Conceptual Sensitivity: Calculated as $ACC_{weak} - ACC_{zero}$. This quantifies Conceptual Sensitivity—how much the model benefits from a simple cultural definition (e.g., the concept of "Mianzi").
- Strong-Weak Gain: Calculated as $ACC_{strong} - ACC_{weak}$. Measures the model's Information Integration Capacity—its ability to move beyond definitions and apply complex cultural logic/reasoning chains.
- Strong-Zero Gain: Calculated as $ACC_{strong} - ACC_{zero}$. Measures the overall improvement in reasoning when explicit cultural context is provided.

2.2 Implementation and Quality Control

2.2.1 Data Dehydration and Automated Pipeline

To prevent a "Ceiling Effect" caused by emotional cues in prompts, this study conducted strict "dehydration processing" on the 50 scenarios during implementation. Subjective adjectives were stripped away, retaining only objective behaviors and power dynamics. Data generation relied on an automated pipeline developed in Python. To address API concurrency limits and network fluctuations, auto-sleep and exponential backoff retry mechanisms were embedded in the collection script, ensuring the integrity of a total of 600 test cases ($50 \text{ scenarios} \times 4 \text{ models} \times 3 \text{ injection strategies}$).

2.2.2 Double-Blind Independent Coding

To maximally eliminate the "Expectancy Effect" in manual evaluation, this study adopted a strict double-blind coding mechanism. All generated responses were mixed and randomized, with their corresponding "model name" and "injection strategy" labels concealed. Two evaluators independently scored the responses without communication, strictly adhering to a highly granular, pre-defined Codebook. Subsequently, the inter-rater reliability (Cohen's Kappa) between the two evaluators was calculated. Discrepancies were resolved through negotiation to finalize the standard score table (Ground Truth) used for calculating accuracy values.

2.2.3 Data Analysis

Data analysis will be performed using Python (Pandas/NumPy) to ensure statistical rigor. We will use visualization tools (Seaborn/Matplotlib) to generate grouped bar charts and line charts, comparing the performance differences and gain trajectories between the Western benchmark model (GPT-4o) and native Chinese models (DeepSeek, Qwen, ERNIE).

3. Result

3.1. Accuracy bar charts for three injection levels

3.1.1 Comparative Analysis of LLM Cultural Intuition: Zero-Injection Baseline

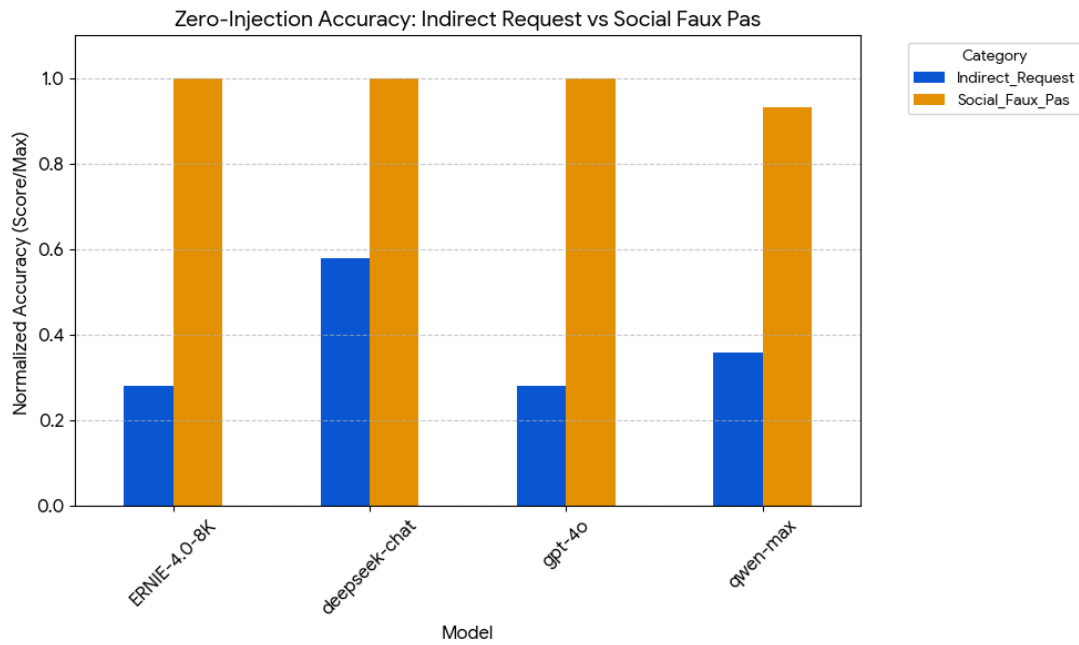


Figure 1 Zero-injection accuracy: indirect request and social faux pas

3.1.2 Comparative Analysis of LLM Cultural Intuition: Weak-Injection

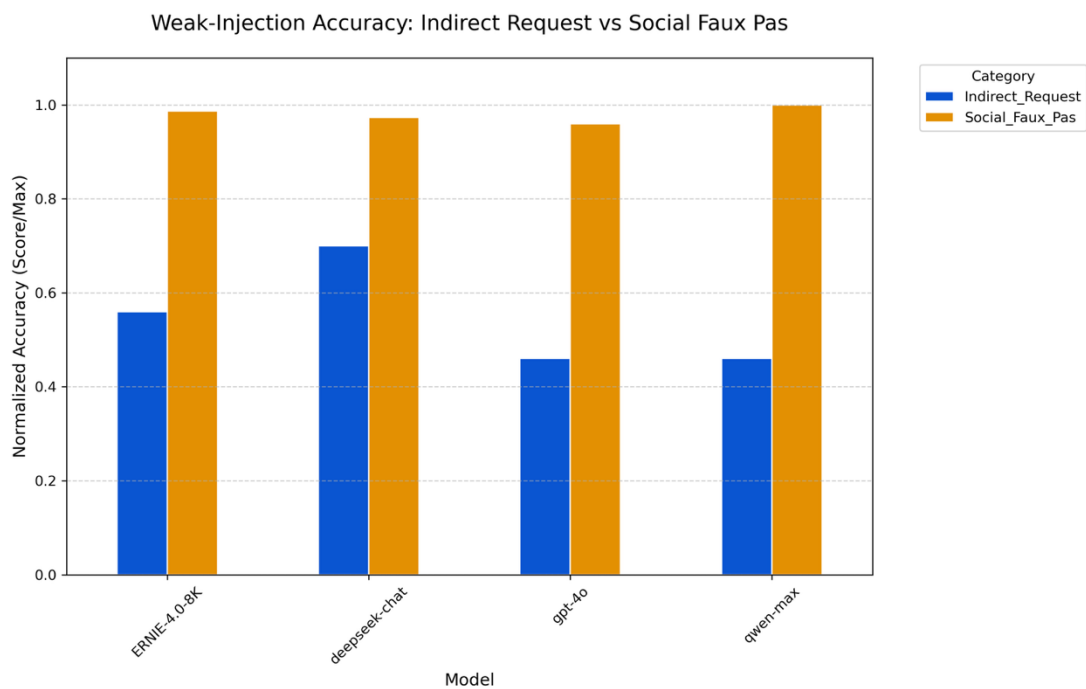


Figure 2 Weak-injection accuracy: indirect request and social faux pas

3.1.3 Comparative Analysis of LLM Cultural Intuition: Strong-Injection

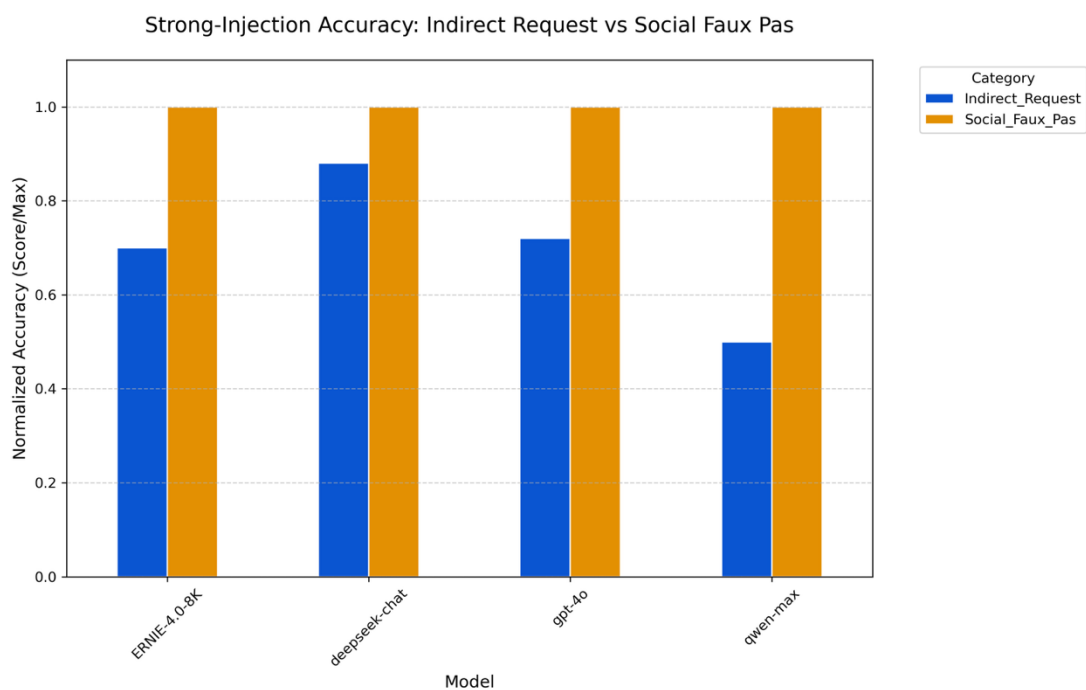


Figure 3 Strong-injection accuracy: indirect request and social faux pas

3.2 Performance Gain Charts

3.2.1 Weak-zero gain

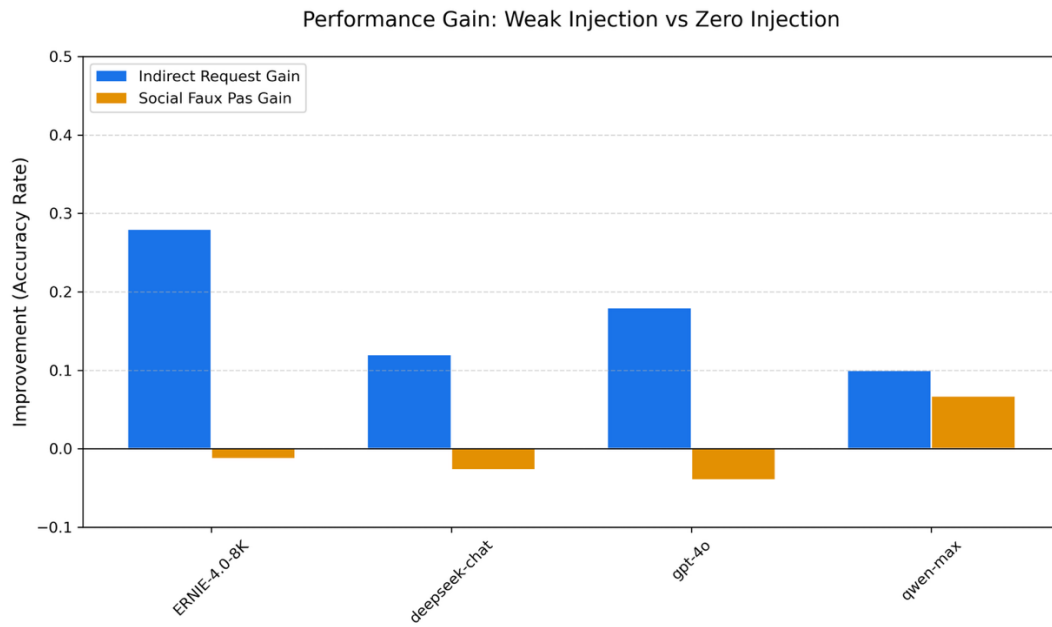


Figure 4 Conceptual Sensitivity

3.2.2 Strong-weak gain

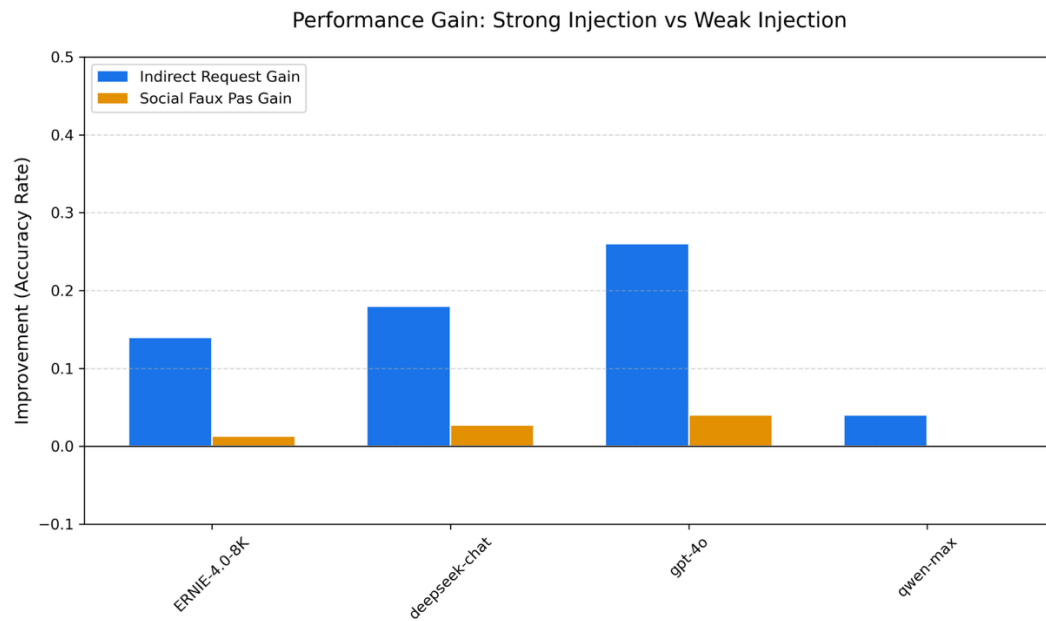


Figure 5 Deep Reasoning Potential

3.2.3 Strong-zero gain

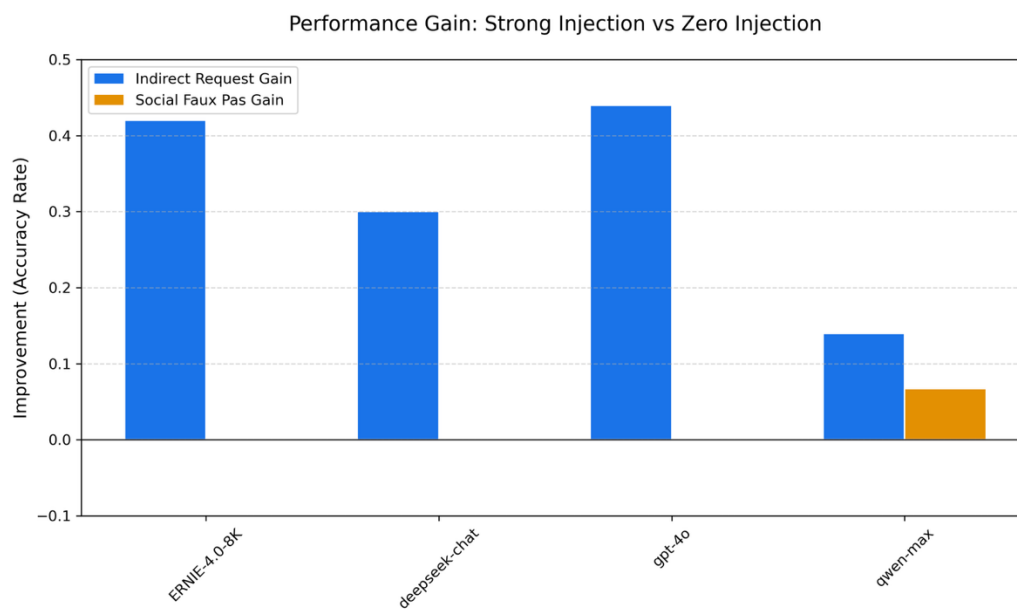


Figure 6 Total Performance Ceiling

3.3 Data analysis

3.3.1 Intrinsic Limitations in Zero-Shot Reasoning

Data reveals a state of "cognitive dissonance" within LLMs when operating within the context of Chinese social dynamics:

All models achieved an accuracy rate of nearly 100% on Task B (Social Faux Pas). This demonstrates that the models have successfully aligned with universally accepted global social boundaries; however, this performance stems more from "pattern matching" against explicit taboos than from a deep-seated cultural understanding.

The average score for Task A (Indirect Requests) was a mere 37.5%, with GPT-4o scoring only 28%. This confirms that, influenced by "Western-centric" training corpora, these models are predisposed to interpreting literal meanings and fail to recognize the "implicit requests"—often conveyed through factual statements—that are characteristic of the Chinese workplace.

DeepSeek-Chat achieved the highest score on Task A (58%), suggesting that, in the absence of explicit prompting, it possesses stronger innate cultural intuition, rather than relying solely on "translational logic" to process Chinese-style subtext.

These benchmarks demonstrate that LLMs generally suffer from a "high-context cognitive impairment." While the models possess cross-cultural knowledge, they are unable to spontaneously map this knowledge onto the complex, real-world social dynamics at play.

Table 1 Zero-injection accuracy

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
DeepSeek-Chat	58.0%	100%
Qwen-Max	36.0%	93.3%
GPT-4o	28.0%	100%
ERNIE-4.0	28.0%	100%

3.3.2 Weak Injection: Conceptual Activation

Following the introduction of definitions for cultural concepts (such as "Mianzi" or "Face"), the models demonstrated a distinct activation effect, yet simultaneously exposed certain cognitive limitations:

ERNIE-4.0 exhibited the strongest conceptual sensitivity, achieving a 28% improvement in Task A and seeing its accuracy rate leap to 56%. This suggests that deep-seated cultural knowledge lies dormant within this domestically developed model, requiring only a simple "label injection" to significantly activate its reasoning pathways. GPT-4o also demonstrated a robust 18% improvement, proving that even Western-centric models can—through the external

injection of Cultural Conceptualizations (CCCs)—partially bridge the cognitive gap inherent in high-context scenarios.

A noteworthy anomaly was observed in Task B: with the exception of Qwen, all other models experienced a negative growth ranging from 1.3% to 4.0%. This phenomenon reflects the "alignment instability" identified by Tao et al. (2024). When specific cultural concepts are forcibly injected, they may interfere with the model's otherwise stable foundation of general social common sense. In over-interpreting the logic of "Mianzi," the model may inadvertently neglect basic social etiquette, resulting in a form of cognitive overload.

Although DeepSeek's relative Gain (12%) was not the highest, its absolute "Weak-injection Accuracy" (70%) remained firmly in first place. This once again validates the power of its native cultural: it requires no drastic logical fluctuations to maintain a leading position within high-context environments.

Weak Injection demonstrates that simple definitions are sufficient to enhance cognitive performance, yet insufficient to achieve full alignment. The slight regression observed in Task B serves as a reminder that mere conceptual injection (Weak Injection) is inherently unstable; it necessitates deeper reasoning guidance (Strong Injection) to firmly anchor the underlying logic.

Table 2 Weak-injection accuracy

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
DeepSeek-Chat	70.0%	97.3%
ERNIE-4.0-8K	56.0%	98.7%
Qwen-Max	46.0%	100.0%
GPT-4o	46.0%	96.0%

Table 3 Accuracy gain from zero-injection to weak-injection

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
ERNIE-4.0-8K	+28.0%	-1.3%
GPT-4o	+18.0%	-4.0%
DeepSeek-Chat	+12.0%	-2.7%
Qwen-Max	+10.0%	+6.7%

3.3.3 Strong Injection: Deep Reasoning and Cultural Alignment

When provided with detailed chains of reasoning and cultural context (referred to as "Strong Injection"), the models' performance evolved from mere "concept activation" to "logical assimilation."

DeepSeek-Chat took first place in Task A with an accuracy rate of 88%. This demonstrates that, building upon a foundation of robust, natively-rooted cultural pre-training—and further bolstered by explicit logical guidance—a model can achieve near-human proficiency in high-context social reasoning tasks. GPT-4o (72%) and ERNIE-4.0 (70%) followed closely behind; their performance indicates that, under conditions of intensive knowledge intervention, even Western-centric models can overcome the limitations of "translated logic" to successfully construct accurate mental maps of Chinese social dynamics.

GPT-4o exhibited the most substantial "Strong-Weak Gain" (+26%). This significant performance leap suggests that GPT-4o possesses exceptional general-purpose logical processing capabilities; although it initially lacks innate Chinese social intuition (as evidenced by its lower scores in the Zero/Weak Injection stages), once provided with comprehensive logical explanations, it can rapidly integrate this information to achieve a level of cross-cultural cognitive transcendence. In contrast, Qwen-Max's gain was a mere +4%, and its final score (50%) placed it at the bottom of the rankings. This outcome reveals a potential "reasoning bottleneck" within the model—specifically, an inability to effectively translate additional background knowledge into accurate social decision-making.

Across the transition from the Weak to the Strong Injection stage, all models that had previously experienced a decline in performance during the Weak stage (e.g., GPT-4o with +4.0% and DeepSeek with +2.7%) demonstrated a full recovery. This outcome validates the critical role played by detailed chains of reasoning. Such reasoning eliminates the "cognitive interference" introduced during the Weak Injection stage, thereby utilizing logical closure to re-stabilize the models' judgment regarding fundamental social etiquette (Task B) and achieving a harmonious synthesis of "cultural depth" and "universal common sense."

The "Strong-Weak Gain" serves as a core metric for assessing a model's capacity for information integration. The results indicate that while GPT-4o may lack innate cultural intuition, it possesses good logical assimilation capabilities; conversely, DeepSeek has demonstrated that the combination of "native cultural DNA" and "explicit logical guidance" constitutes the optimal pathway toward achieving perfect cultural alignment.

Table 4 Strong-injection accuracy

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
DeepSeek-Chat	88.0%	100.0%
GPT-4o	72.0%	100.0%
ERNIE-4.0-8K	70.0%	100.0%

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
Qwen-Max	50.0%	100.0%

Table 5 Accuracy gain from Strong-injection to weak-injection

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
GPT-4o	+26.0%	+4.0%
DeepSeek-Chat	+18.0%	+2.7%
ERNIE-4.0-8K	+14.0%	+1.3%
Qwen-Max	+4.0%	+0.0%

3.3.4 Strong-Zero Gain

The Strong-Zero Gain metric measures a model's capacity for adaptation, and its "upper bound for cross-cultural alignment" under conditions of extreme contextual pressure.

GPT-4o (+44%) and ERNIE-4.0 (+42%) demonstrated astonishing total gains. This indicates that even if a model is initially extremely insensitive to Chinese-style "subtext" in its native

state (achieving only 28% accuracy), providing a complete cultural-logical framework enables it to undergo a qualitative transformation. This also validates the hypothesis proposed by Tao et al. (2024): that models can, through logical compensation, partially overcome the Western-centric biases inherent in their pre-training corpora.

DeepSeek-Chat's (+30%) total gain appears lower than that of the GPT models; however, when viewed in conjunction with its final accuracy rate of 88%, it represents a distinct pattern: a high starting point combined with a high ceiling. Its 30% improvement demonstrated a synergy between its intrinsic foundation of native cultural corpora (Intrinsic Patterns) and the injection of external logic (Extrinsic Logic).

GPT, ERNIE, and DeepSeek all recorded a 0.0% gain on Task B. This does not imply a lack of progress; rather, it is because these models had already achieved 100% alignment with moral standards during the zero-injection stage. The strong injection merely served to reinforce this understanding, ensuring that fundamental common sense regarding social etiquette would not be lost during complex cultural analyses.

Qwen-Max's total gain on Task A was a mere 14%; even with the strongest external assistance, it lagged significantly behind the other models. This suggests that the model suffers from significant cognitive rigidity in tasks requiring high-context social reasoning, making it difficult to activate its latent "Theory of Mind" (ToM) capabilities through prompt engineering. Based on the evolutionary analysis across these four stages, we can draw the following core conclusion: the existence of "high-context blind spots" is a universal phenomenon. Unaligned LLMs are generally incapable of processing Chinese-style "subtext," thereby providing

empirical evidence for the tangible existence of Western-centric bias. With its Strong-Zero gains, GPT-4o has demonstrated that even in the absence of native cultural intuition, the injection of deep logical chains can effectively construct accurate cultural mappings. Meanwhile, DeepSeek’s consistently superior accuracy proves that "native training data" establishes a higher ceiling for reasoning capabilities when handling Chinese social dynamics.

Table 6 Accuracy gain from Strong-injection to zero-injection

Model	Accuracy of Indirect request	Accuracy of social Faux Pas
GPT-4o	+44.0%	+0.0%
ERNIE-4.0-8K	+42.0%	+0.0%
DeepSeek-Chat	+30.0%	+0.0%
Qwen-Max	+14.0%	+6.7%

4. Discussion

The results of this study reveal the complex interplay between latent cultural biases and the capacity for cultural alignment in Large Language Models (LLMs). Our data confirms a "high-context cognitive impairment" in default states; while models align with universal social taboos

(Task B), they struggle significantly with the implicit logic of Chinese indirect requests (Task A), with GPT-4o scoring only 28%. This aligns with the findings of Kharchenko et al. (2025), who argued that LLMs inherently mirror the Western-centric values of their dominant English training corpora. However, our experimental findings demonstrate the immense potential of "Cultural Prompting" in enhancing the localisation of globalised AI services. The "Strong-Zero Gain" observed in our study (Figure 6)—where GPT-4o improved by 44%—demonstrates that "logical compensation" can partially bridge this gap. This supports the assertion by Tao et al. (2024) that models can function as "cultural simulators" when provided with explicit reasoning chains. Yet, the observed "alignment instability" during weak injection suggests that mere conceptual labels without deep logic can destabilize a model's existing social common sense.

4.1 Potential Applications

As noted by Kharchenko et al. (2025), by utilizing context-specific prompts, models can provide more culturally adaptive recommendations for groups currently underrepresented in mainstream technology—such as those within the Afro-Islamic or Western European cultural spheres. This capability is particularly critical for educational and professional writing in non-Western settings, as it effectively prevents users from being compelled to assimilate into dominant Western "self-expression" values during AI-assisted communication, thereby preserving their original cultural authenticity. Furthermore, conceptualising LLMs as "compression algorithms for human culture" carries profound educational implications; Tao et al. (2024) argue that this framework allows models to function as cultural simulators, enabling users to explore heterogeneous social norms within simulated advice-seeking scenarios.

4.2 Limitations and Future Direction

Despite these findings, several limitations must be acknowledged. First, the subjectivity of scoring represents a primary constraint; evaluations were conducted by group members rather than a diverse panel of experts, which may introduce individual cognitive bias. As Tao et al. (2024) noted, capturing deep cultural nuances requires authoritative validation to ensure psychological accuracy. Second, the diversity of models was limited; using only GPT-4o to represent Western AI and a small selection of Chinese models may not capture the full spectrum of cross-cultural performance across different architectures. Finally, the sample size of 50 scenarios, while representative, is insufficient to capture the full granularity of Chinese social dynamics, where biases often hide in low-frequency interactions.

To address these gaps, future steps will involve scaling the dataset to over 500 scenarios and implementing expert-led evaluation and crowdsourced validation to ensure the "Face-saving" scores reflect broader societal norms. Additionally, we plan to expand the model matrix to include Llama 3 and Claude to further verify the universality of the "English-Lens" bias. Furthermore, future research endeavors should focus on moving beyond the confines of mere prompt engineering. Incorporating Retrieval-Augmented Generation (RAG) techniques—thereby integrating empirically validated external benchmarks such as the World Values Survey (WVS) or Hofstede's cultural dimensions into the generative process—may effectively mitigate models' reliance on stereotypes. As Kharchenko et al. (2025) emphasise, merely adjusting prompts is insufficient; future development must instead deeply embed diverse ethical norms by fine-tuning models on more representative, localized datasets. Finally, the

academic community urgently needs to address the long-term societal implications of cultural bias in AI—specifically, whether such bias might, over time, subtly reshape users' own cultural attitudes and values.

References:

Guo, S., Jiang, S., He, Q., Xiao, Y., Liang, J., Yude, B., ... & Zhang, L. (2025). Do Large Language Models Truly Understand Cross-cultural Differences?.arXiv preprint arXiv:2512.07075.

Kharchenko, J., Roosta, T., Chadha, A., & Shah, C. (2024). How well do llms represent values across cultures? empirical analysis of llm responses based on hofstede cultural dimensions.arXiv preprint arXiv:2406.14805.

Strachan, J. W., Albergo, D., Borghini, G., Pansardi, O., Scaliti, E., Gupta, S., ... & Becchio, C. (2024). Testing theory of mind in large language models and humans.Nature human behaviour,8(7), 1285-1295.

Tao, Y., Viberg, O., Baker, R. S., & Kizilcec, R. F. (2024). Cultural bias and cultural alignment of large language models.PNAS nexus,3(9), pgae346.